Exercises learning task 6

2026 – 2027

Exercise 4.51: Assume that Z has a standard normal distribution. Determine the value for z that solves each of the following:

(a) $P(Z < z) = 0.9$
(b) $P(Z < z) = 0.5$
(c) $P(Z > z) = 0.1$
(d) $P(Z > z) = 0.9$
(e) $P(-1.28 < Z < z) = 0.8$

Exercise 4.55: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the following:

(a) $P(X < 11)$
(b) $P(X > 2)$
(c) $P(3 < X < 7)$
(d) $P(-2 < X < 9)$
(e) $P(2 < X < 8)$

Exercise 4.56: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the value for x that solves each of the following:

(a) $P(X > x) = 0.5$
(b) $P(X > x) = 0.95$
(c) $P(x < X < 7) = 0.2$
(d) $P(3 < X < x) = 0.95$
(e) $P(-x < X - 5 < x) = 0.99$

Exercise 4.63: The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of x cc and a standard deviation of 5 cc.

(a) At what value should the mean be set so that $99.9\,\%$ of all cans exceed 365 cc?

(b) At what value should the mean be set so that $99.9\,\%$ of all cans exceed 365 cc if the standard deviation can be reduced to 2 cc?

Exercise 4.69: The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.

(a) What is the probability that a laser fails before 6000 hours?

(b) What is the life in hours that $95\,\%$ of the lasers exceed?

(c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

Exercise 4.91: Suppose X has an exponential distribution with mean equal to 10. Determine the following:

(a) $P(X > 10)$
(b) $P(X > 20)$
(c) $P(X < 30)$
(d) Find the value of x such that $P(X < x) = 0.90$.

Exercise 4.103: The lifetime of a mechanical assembly in a vibration test is exponentially distributed with a mean of 400 hours.

(a) What is the probability that an assembly on test fails in less than 200 hours?

(b) What is the probability that an assembly operates for more than 500 hours before failure?

(c) If an assembly has been on test for 400 hours without a failure, what is the probability of a failure in the next 100 hours?

(d) If 10 assemblies are tested, what is the probability that at least one fails in less than 100 hours? Assume that the assemblies fail independently.

(e) If 10 assemblies are tested, what is the probability that all have failed by 800 hours? Assume that the assemblies fail independently.

Exercise 4.165: Suppose that $f(x) = 0.5x - 1$ for $2 < x < 4$. Determine the following:

(a) $P(X < 2.5)$ (b) $P(X > 3)$ (c) $P(2.5 < X < 3.5)$
(d) Determine the cumulative distribution function of the random variable.
(e) Determine the mean and variance of the random variable.